A Guideline-Informed AI Agent

for Empiric Antibiotic Decision Support in the ICU

[Team 5] Minso Kim, Soyoung Park, Taewon Jung, Jisu Ha, Seungmin Lee, Woojin Kim

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Members

3 Technicians ∩ 3 Clinicians

“It runs fine on my GPU.”

“…but at 3 a.m. in the ICU?”

Technicians

Minso Kim

Soyoung Park

Taewon Jung

Team 5

{ }

Clinicians

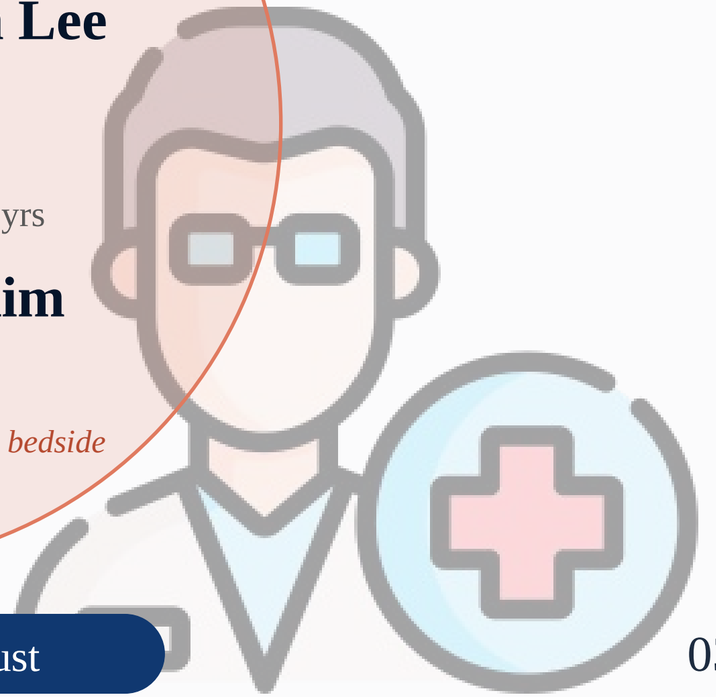
Seungmin Lee
Neurologist

Jisu Ha
ICU nurse, > 10 yrs

Woojin Kim
ICU nurse

sepsis · antibiotics · guidelines · bedside





Code × Care = an antibiotic AI agent clinicians may trust

Background

Empirical Antibiotic Therapy

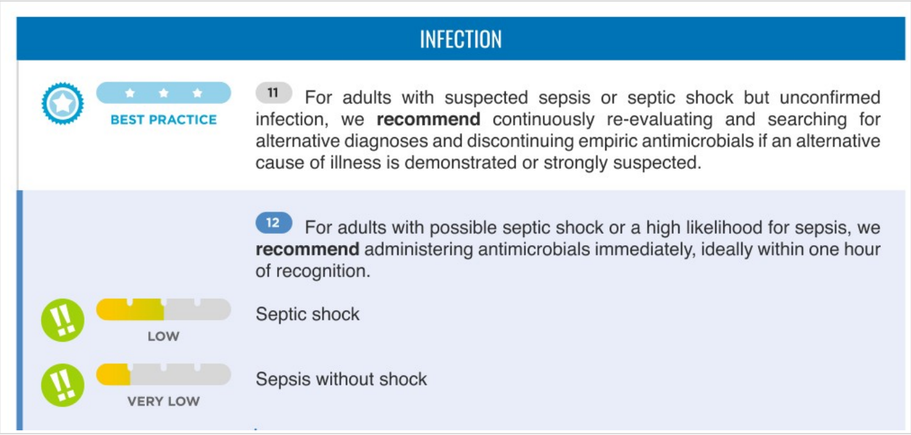
Definition [1]

Antibiotics started **before the pathogen is identified**

pathogen unknown → *pathogen identified*

Suspected infection → **Empiric antibiotics** → **Culture results** → **Targeted / de-escalation**

Surviving Sepsis Campaign 2021 [2]



INFECTION

BEST PRACTICE

11 For adults with suspected sepsis or septic shock but unconfirmed infection, we **recommend** continuously re-evaluating and searching for alternative diagnoses and discontinuing empiric antimicrobials if an alternative cause of illness is demonstrated or strongly suspected.

12 For adults with possible septic shock or a high likelihood for sepsis, we **recommend** administering antimicrobials immediately, ideally within one hour of recognition.

LOW — Septic shock

VERY LOW — Sepsis without shock

Table 1 (excerpt), Evans et al., Intensive Care Med 2021 [2]

Why it matters [2, 3]

≤ 1 h

antimicrobials for possible septic shock / high likelihood of sepsis

1. **Timely** — early treatment in severe infection
2. **Appropriate** — source · pathogen · severity
3. **Not excessive** — adverse events · resistance
4. **Reassessed** — de-escalate with cultures

High-stakes decision under uncertainty

→ **guideline-concordant choice matters**

[1] Rhodes A, Evans LE, Alhazzani W, Levy MM, Antonelli M, Ferrer R, et al. Surviving Sepsis Campaign: international guidelines for management of sepsis and septic shock: 2016. Intensive Care Med. 2017;43(3):304-77. doi:10.1007/s00134-017-4683
[2] Evans L, Rhodes A, Alhazzani W, Antonelli M, Coopersmith CM, French C, et al. Surviving Sepsis Campaign: international guidelines for management of sepsis and septic shock 2021. Intensive Care Med. 2021;47(11):1181-247. doi:10.1007/s00134-021-06506-y. Table 1 (excerpt), CC BY-NC
[3] Tamma PD, Bonomo RA, Heil EL, Justo JA, Satlin MJ, Mathers AJ. Infectious Diseases Society of America 2026 guidance on the treatment of antimicrobial-resistant gram-negative infections. Clin Infect Dis. 2026. doi:10.1093/cid/ciag4

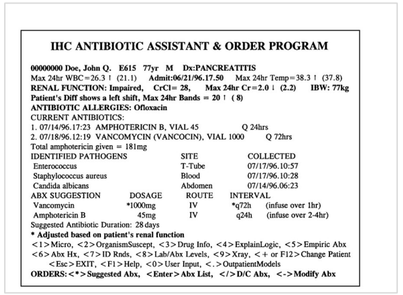
Background

Previous Studies

Roles of CDSS [4]

- **Prescribing support**
- **Guideline implementation**
- **Dose optimization**
- **Use monitoring**

Rule-based CDSS 1998 – 2024



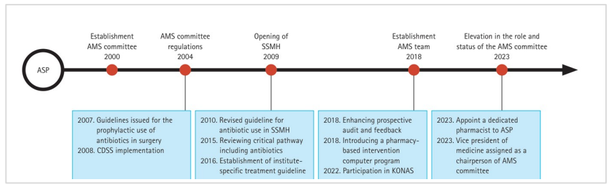
IHC ANTIBIOTIC ASSISTANT & ORDER PROGRAM

00000000 Doe, John Q. E615 77yr M Dx:PANCREATITIS
Max 24hr WBC=26.3 ↑ (21.1) Admit:06/21/96.17.50 Max 24hr Temp=38.3 ↑ (37.8)
RENAL FUNCTION: Impaired, CrCl= 28, Max 24hr Cr=2.0 ↑ (2.2) IBW: 77kg
Patient's Diff shows a left shift, Max 24hr Bands = 20 ↑ (8)
ANTIBIOTIC ALLERGIES: Ofloxacin
CURRENT ANTIBIOTICS:
1. 07/14/96.17:23 AMPHOTERICIN B, VIAL 45 Q 24hrs
2. 07/18/96.12:19 VANCOMYCIN (VANCOCIN), VIAL 1000 Q 72hrs
Total amphotericin given = 181mg
IDENTIFIED PATHOGENS SITE COLLECTED
Enterococcus T-Tube 07/17/96.10:57
Staphylococcus aureus Blood 07/17/96.10:28
Candida albicans Abdomen 07/14/96.06:23
ABX SUGGESTION DOSAGE ROUTE INTERVAL
Vancomycin *1000mg IV *q72h (infuse over 1hr)
Amphotericin B 45mg IV q24h (infuse over 2-4hr)
Suggested Antibiotic Duration: 28 days
* Adjusted based on patient's renal function
<1> Micro, <2> OrganismSuscept, <3> Drug Info, <4> ExplainLogic, <5> Empiric Abx
<6> Abx Hx, <7> ID Rsdt, <8> Lab/Abx Levels, <9> Xray, < + or F12> Change Patient
<Esc> EXIT, <F1> Help, <0> User Input, <.> OutpatientModels
ORDERS: <*> Suggested Abx, <Enter> Abx List, </> D/C Abx, <-> Modify Abx

Evans et al. (USA, 1998) [5]

28 → 4

anti-infective adverse events with a computerized ICU program

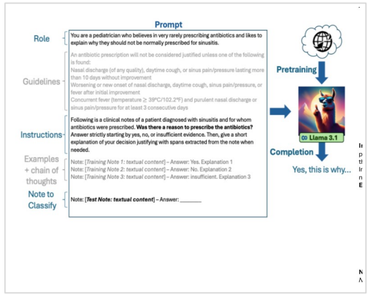


Lee et al. (Korea, 2024) [6]

CDSS since 2008

stepwise stewardship in a Korean tertiary hospital

Generative AI / LLM 2025 – 2026



Weissenbacher et al. (2025) [7]

94.7% / 66.2%

appropriate / inappropriate prescriptions identified (pediatric sinusitis)

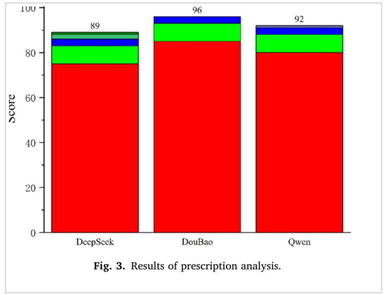


Fig. 3. Results of prescription analysis.

Zhang et al. (2026) [8]

3 LLMs · 20 cases

real-world outpatient prescriptions with problems

Gap: single condition or small, preselected samples

[4] Barlam TF, Cosgrove SE, Abbo LM, MacDougall C, Schuetz AN, Septimus EJ, et al. Implementing an antibiotic stewardship program: guidelines by the Infectious Diseases Society of America and the Society for Healthcare Epidemiology of America. Clin Infect Dis. 2016;62(10):e51-77. doi:10.1093/cid/ciw
[5] Evans RS, Pestotnik SL, Classen DC, Clemmer TP, Weaver LK, Orme JF Jr, et al. A computer-assisted management program for antibiotics and other antiinfective agents. N Engl J Med. 1998;338(4):232-8. doi:10.1056/NEJM199801223380406. Figur
[6] Lee SJ, Lee R, Cho SY, Nho D, Ahn HL, Lee DG. Implementing antimicrobial stewardship: lessons and perspectives from a university-affiliated tertiary hospital in Korea. Korean J Intern Med. 2024;39(3):399-412. doi:10.3904/kjim.2023.504. Figure 1, CC BY-NC
[7] Weissenbacher D, Dutcher L, Boustany M, Cressman L, O'Connor K, Hamilton KW, et al. Automated evaluation of antibiotic prescribing guideline concordance in pediatric sinusitis clinical notes. Pac Symp Biocomput. 2025;30:138-53. doi:10.1142/9789819807024_0011. Fig. 1 (
[8] Zhang W, Xu J, Dong T, Hao X, Yang Q, Zhang J, et al. Exploring large language models as a prescription decision support tool for rational antibiotic use: a dual-framework analysis using standardized examinations and real-world clinical cases. Explor Res Clin Soc Pharm. 2026;23:100821. doi:10.1016/j.rcsop.2026.100821. Fi

Background

Why AI agents? — Our Approach (novelty)

AI agent = LLM + tools · retrieval · memory [9]

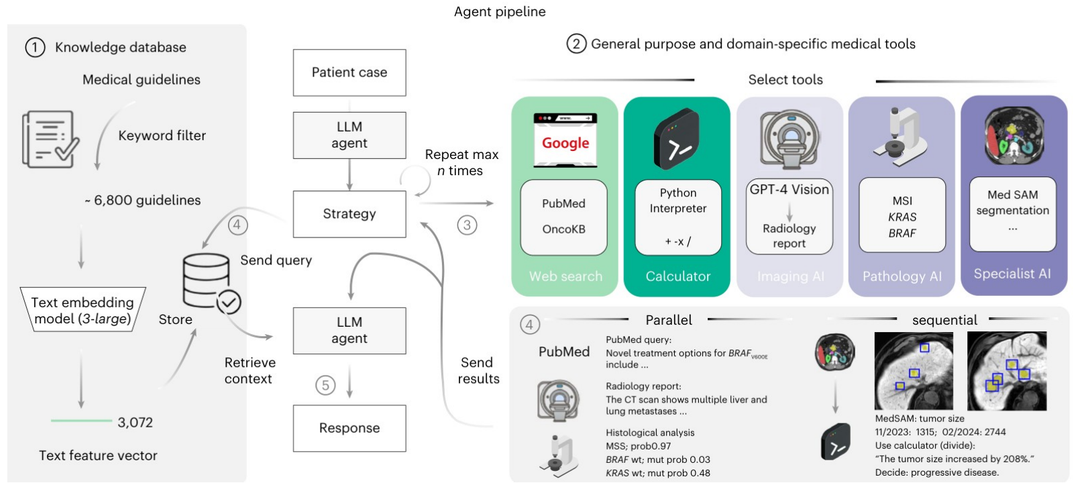


Fig. 1, Ferber et al., Nat Cancer 2025 [10]

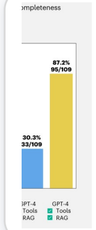
	LLM	Traditional agent	LLM-based agent
Mastery of language			
High-level cognitive abilities such as reasoning and reflection			
Long-term memory			
Open-ended action space			
Generalizable and not task specific			
Trustworthy outputs			
Tool-using and collaborative skills			
Less dependent on human prompts or instructions			
Behaviour is predictable			
Can interact with multimodal inputs			

Fig. 1, Qiu et al., Nat Mach Intell 2024 [9]

LLM-based agents

- memory
- tool use
- open-ended actions
- multimodal input

Gap: one condition · small samples · one guideline set



30.3% → 87.2%

GPT-4 alone → agent with tools + RAG [10]

up to +18%

accuracy with medical RAG

6 LLMs · 7,663 questions [11]

This study (novelty)

Pneumonia & UTI (± sepsis)

MIMIC-IV (US) + K-MIMIC (Korea)

National + WHO guidelines

3-function agent · clinician-evaluated

[9] Qiu J, Lam K, Li G, Acharya A, Wong TY, Darzi A, et al. LLM-based agentic systems in medicine and healthcare. Nat Mach Intell. 2024;6(12):1418-20. doi:10.1038/s42256-024-00944-1. Fig.
[10] Ferber D, El Nahhas OSM, Wölflein G, Wiest IC, Clusmann J, Leßmann ME, et al. Development and validation of an autonomous artificial intelligence agent for clinical decision-making in oncology. Nat Cancer. 2025;6(8):1337-49. doi:10.1038/s43018-025-00991-6. Fig. 1, Fig. 2b, CC BY
[11] Xiong G, Jin Q, Lu Z, Zhang A. Benchmarking retrieval-augmented generation for medicine. In: Findings of the Association for Computational Linguistics: ACL 2024. Bangkok: ACL; 2024. p. 6233-51. doi:10.18653/v1/2024.findings-acl.3

Data

MIMIC-IV & K-MIMIC

MIMIC-IV

▼ narrowed down to

Pneumonia

with sepsis

Urinary Track Infection

with sepsis

K-MIMIC

▼ narrowed down to

Pneumonia

Urinary Track Infection

Data

Guidelines

Korean guidelines → K-MIMIC

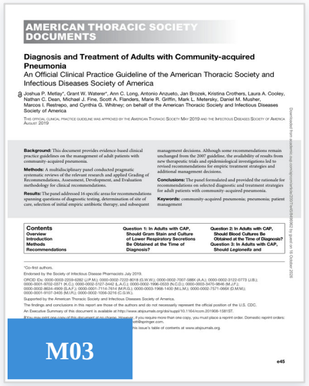


K02

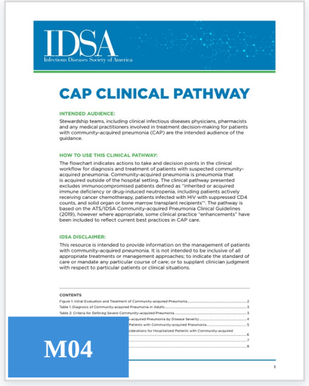


K04

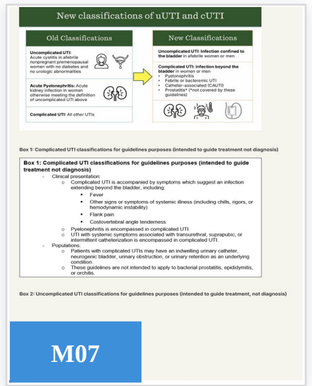
USA guidelines → MIMIC-IV



M03

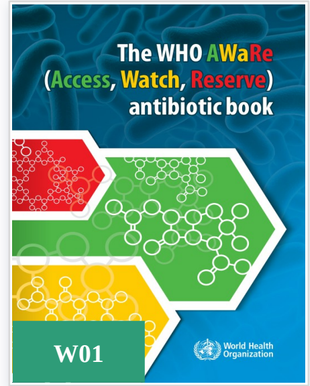


M04

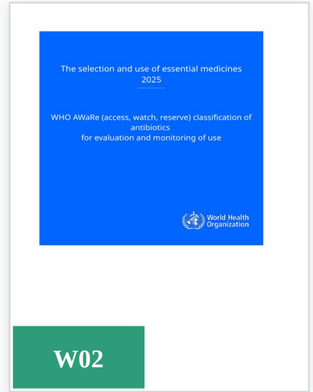


M07

International (WHO) → both



W01



W02

	Guideline	Issuer, year	Infection	K-MIMIC	MIMIC-IV
K02	Practical guidelines for appropriate antibiotic use in community-acquired pneumonia [12]	KDCA · KSAT, 2025	Pneumonia	●	–
K04	Practical guidelines for appropriate antibiotic use in urinary tract infection [13]	KDCA · KSAT, 2025	UTI	●	–
M03	Diagnosis and treatment of adults with community-acquired pneumonia [14]	ATS / IDSA, 2019	Pneumonia	–	●
M04	CAP clinical pathway (antimicrobial stewardship) [15]	IDSA	Pneumonia	–	●
M07	Guideline on the management of complicated urinary tract infections [16]	IDSA, 2025	UTI	–	●
W01	The WHO AWaRe (Access, Watch, Reserve) antibiotic book [17]	WHO, 2022	Pneumonia · UTI	●	●
W02	WHO AWaRe classification of antibiotics for evaluation and monitoring of use [18]	WHO, 2025	All antibiotics	●	●

[12] Korea Disease Control and Prevention Agency, Korean Society for Antimicrobial Therapy. Practical guidelines for appropriate antibiotic use in community-acquired pneumonia (2025 ed.). Cheongju: KDCA; 2026. Available from: https://www.kdca.go.kr/bbs/kdca/55/310916/artclView
[13] Korea Disease Control and Prevention Agency, Korean Society for Antimicrobial Therapy. Practical guidelines for appropriate antibiotic use in urinary tract infection (2025 ed.). Cheongju: KDCA; 2026. Available from: https://www.kdca.go.kr/bbs/kdca/55/310916/artclView
[14] Metlay JP, Waterer GW, Long AC, Anzueto A, Brozek J, Crothers K, et al. Diagnosis and treatment of adults with community-acquired pneumonia: an official clinical practice guideline of the ATS and IDSA. Am J Respir Crit Care Med. 2019;200(7):e45-e67. doi:10.1164/rccm.201908-1581
[15] Infectious Diseases Society of America. Community-acquired pneumonia (CAP) clinical pathway. Arlington (VA): IDSA; 2023. Available from: https://www.idsociety.org/practice-guideline/community-acquired-pneumonia-cap-in-adu
[16] Trautner BW, Cortés-Penfield NW, Gupta K, Hirsch EB, Horstman M, Moran GJ, et al. IDSA 2025 guideline on management and treatment of complicated urinary tract infections: selection of antibiotic therapy. Clin Infect Dis. 2026;82(Suppl 3):i36-i67. doi:10.1093/cid/ciaf
[17] World Health Organization. The WHO AWaRe (Access, Watch, Reserve) antibiotic book. Geneva: WHO; 2022. ISBN 978-92-4-006238-2. Available from: https://www.who.int/publications/i/item/97892400623
[18] World Health Organization. The selection and use of essential medicines, 2025: WHO AWaRe (Access, Watch, Reserve) classification of antibiotics for evaluation and monitoring of use. Geneva: WHO; 2025. doi:10.2471/B094

Method

Study Population Filtration (1/3)

A. Cohort construction (MIMIC-IV v3.1; counts = admissions)

MIMIC-IV v3.1 (hosp + icu modules)
364,627 patients · 546,028 admissions · 94,458 ICU stays

First ICU stay per patient · adults ≥ 18 years
65,366 patients, one admission each (readmissions excluded)

Diagnosis codes, primary + secondary (ICD-9 / ICD-10)

Pneumonia 8,234
ICD-10: J13–J16, J18.0/.1/.8/.9
ICD-9: 481–483, 485, 486

UTI 7,793
ICD-10: N10, N12, N39.0
ICD-9: 590.1/.80/.9, 599.0

Excluded
Aspiration 815 · VAP 419 (pneumonia)
Catheter-associated 330 (UTI)
Both pneumonia & UTI codes 1,029

After exclusion
Pneumonia 6,056 · UTI 6,434

Suspected-infection time = culture ↔ antibiotic pairing
culture_first (culture → antibiotic ≤ 72 h): 4,752 · 5,464
antibiotic_first (antibiotic → culture ≤ 24 h): 1,009 · 628
Suspicion time assigned: 5,761 · 6,092

No culture–antibiotic pair: 295 · 342

Hospital-acquired (> 48 h): 886 · 1,569

Community-acquired: suspicion ≤ 48 h after hospital arrival
Pneumonia 4,875 · UTI 4,523 (final cohort)

Definitions

Sepsis-3: suspected infection + acute organ dysfunction (SOFA ≥ 2) [19]

Suspected infection: culture & antibiotic within 72 h / 24 h of each other [20]

Sepsis classification

three labels (pneumonia · UTI)

	PNA · UTI
Sepsis-3 (mimic-code official; primary label) [21]	**3,666 · 3,180**
Simplified SOFA ≥ 2 (shared code, also applied to K-MIMIC)	**3,858 · 3,496**
ICD sepsis codes	**1,721 · 1,526**

Agreement with Sepsis-3: SOFA 85% · 82% / codes 50% · 54%

18% had no ICU record within the SOFA window → **flagged, not excluded**

[19] Singer M, Deutschman CS, Seymour CW, Shankar-Hari M, Annane D, Bauer M, et al. The third international consensus definitions for sepsis and septic shock (Sepsis-3). JAMA. 2016;315(8):801-10. doi:10.1001/jama.2016.02
[20] Seymour CW, Liu VX, Iwashyna TJ, Brunkhorst FM, Rea TD, Scherag A, et al. Assessment of clinical criteria for sepsis: for the Third International Consensus Definitions for Sepsis and Septic Shock (Sepsis-3). JAMA. 2016;315(8):762-74. doi:10.1001/jama.2016.02
[21] Johnson AE, Stone DJ, Celi LA, Pollard TJ. The MIMIC Code Repository: enabling reproducibility in critical care research. J Am Med Inform Assoc. 2018;25(1):32-9. doi:10.1093/jamia/ocx0

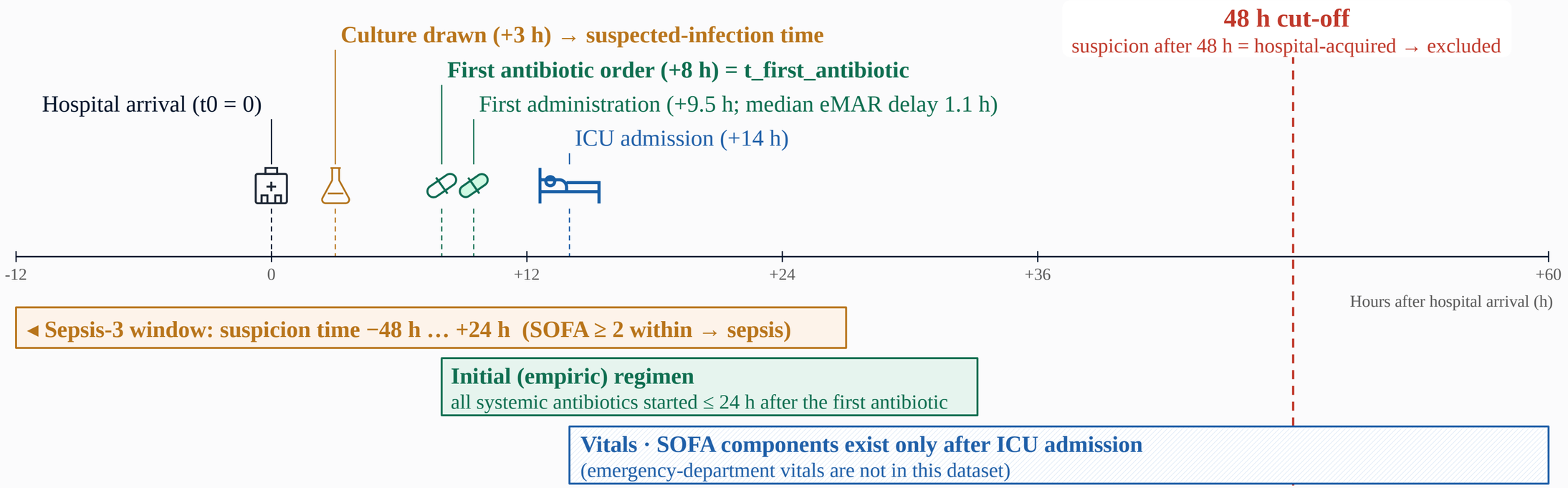
Method

Study Population Filtration (2/3)

B. One patient's timeline and rule windows (hours from hospital arrival, t0 = 0)

Rule 1 · culture_first — Culture → (≤ 72 h) → Antibiotic ⇒ **Suspicion time = culture time**

Rule 2 · antibiotic_first — Antibiotic → (≤ 24 h) → Culture ⇒ **Suspicion time = antibiotic time**



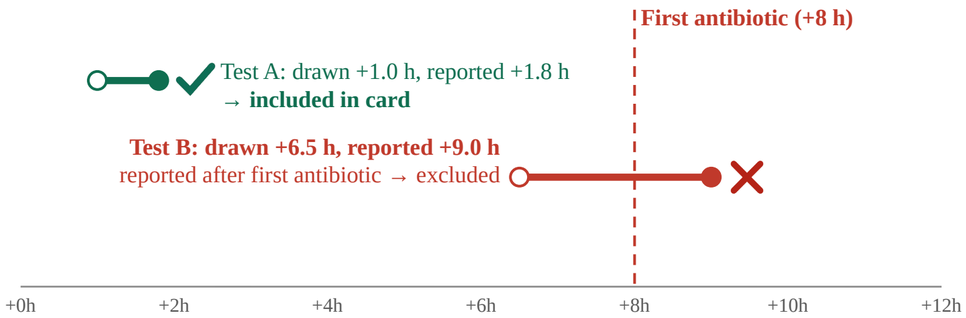
[19] Singer M, Deutschman CS, Seymour CW, Shankar-Hari M, Annane D, Bauer M, et al. The third international consensus definitions for sepsis and septic shock (Sepsis-3). JAMA. 2016;315(8):801-10. doi:10.1001/jama.2016.02
[20] Seymour CW, Liu VX, Iwashyna TJ, Brunkhorst FM, Rea TD, Scherag A, et al. Assessment of clinical criteria for sepsis: for the Third International Consensus Definitions for Sepsis and Septic Shock (Sepsis-3). JAMA. 2016;315(8):762-74. doi:10.1001/jama.2016.02

Method

Study Population Filtration (3/3)

C. Leakage-free case card given to the agent (hashed ID; all times relative to first antibiotic = T+0)

Case-card availability rule
only values whose result was reported (storetime) before the first antibiotic



*Selecting by draw time would leak results the physician had not yet seen into **13% of 'last values' and 21% of patients***

Case cards → LLM agent
hashed ID · relative time

20	20
evaluation cases (5 × 4 groups)	reserve cases

case.json

- Age · sex · weight · ED pathway · ICU type
- Vitals · labs: last value reported before T+0
- Cultures before T+0: specimen type & time only, no results
- Empiric regimen: agent · class · AWaRe · route · dose · start T+h
- History: admission / IV antibiotics ≤ 90 d · MRSA / Pseudomonas ≤ 1 y
- Comorbidities (codes): diabetes, CKD, COPD, immunosuppression, etc.

Not in the card

- ✕ Discharge diagnosis codes
- ✕ In-hospital death
- ✕ Culture results after T+0
- ✕ SOFA score / sepsis label

Judgement

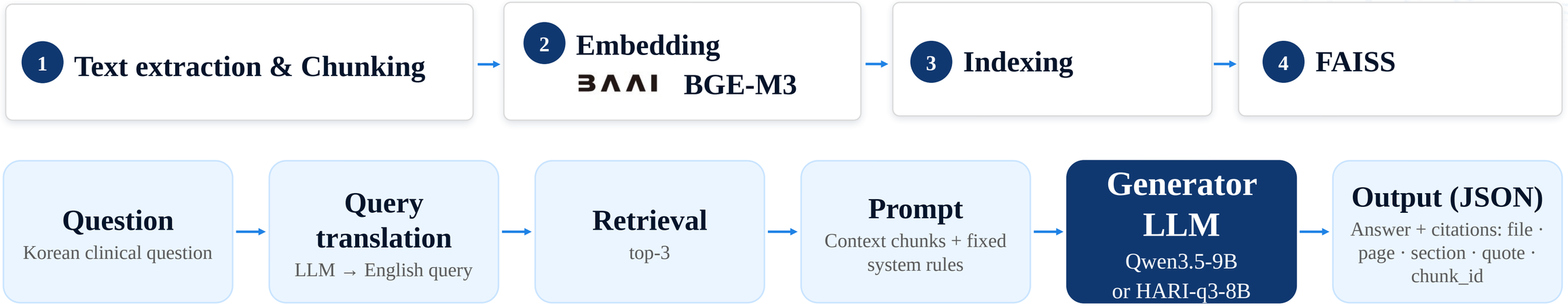
A1 syndrome · severity → A4 domestic / international guideline

Adherence judgement → A5 evidence verification

Clinician review: 5 cases × 4 groups (1–5 score, 4-level verdict)

Method

Embedding Vector Construction & RAG LLM Model Comparison



Qwen3.5-9B

Alibaba Cloud · Qwen team [22]

TYPE

General-purpose multilingual LLM

HARI-q3-8B

Seoul National University Hospital [23]

TYPE

Korean medical LLM built on Qwen3-8B-Base

Identical for both LLMs: same embedding & FAISS index · same 9 retrieved chunks · same prompt · temperature 0.2, seed 42, context 8,192 · local inference

[22] Qwen Team. Qwen3.5: towards native multimodal agents [Internet]. Alibaba Cloud; 2026. Available from: https://qwen.ai/blog?id=qwen3.5 ; Yang A, Li A, Yang B, Zhang B, Hui B, Zheng B, et al. Qwen3 technical report. arXiv:2505.09388. 20
[23] Healthcare AI Research Institute, Seoul National University Hospital. hari-q3-8b [model card]. Hugging Face; 2025. Available from: https://huggingface.co/snuh/hari-q3-

Method

RAG Evaluation

① Automatic metrics vs. reference answer

Embedding cosine similarity [24]

BERTScore F1 [25]

ROUGE-L F1 [26, 27]

Token F1 [28]

② Clinician evaluation 5 criteria × 5-point Likert [29, 30, 31]

ICU nurse
junior

ICU nurse
> 10 years

Neurologist
physician

independent scoring

Criterion	What is assessed	1 point	5 points
Accuracy	agreement with guideline content, dose, conditions	key recommendation wrong / no answer	numbers, conditions & exceptions all correct
Relevance	directly answers the question	unrelated to the question	direct and concise for the whole question
Completeness	essential elements of the reference answer	almost no essential element	all elements, conditions & exceptions
Groundedness	citation location is correct and supports the claim	no / non-existent / unsupportive citation	file, page & location exact; all claims supported
Clinical usefulness	helps interpretation and decision making	dangerous misunderstanding / unusable	clear; conditions and limits made explicit

[24] Chen J, Xiao S, Zhang P, Luo K, Lian D, Liu Z. M3-Embedding: multi-linguality, multi-functionality, multi-granularity text embeddings through self-knowledge distillation. In: Findings of ACL 2024. Bangkok: ACL; 2024. p. 2318-35. doi:10.18653/v1/2024.findings-acl.
[25] Zhang T, Kishore V, Wu F, Weinberger KQ, Artzi Y. BERTScore: evaluating text generation with BERT. In: International Conference on Learning Representations (ICLR); 2020. arXiv:1904.096
[26] Lin CY. ROUGE: a package for automatic evaluation of summaries. In: Text Summarization Branches Out. Barcelona: ACL; 2004. p. 74-8
[27] Lee M. Kiwi: developing a Korean morphological analyzer based on statistical language models and skip-bigram. Korean J Digit Humanit. 2024;1(1):109-36. doi:10.23287/KJDH.2024.1.1
[28] Rajpurkar P, Zhang J, Lopyrev K, Liang P. SQuAD: 100,000+ questions for machine comprehension of text. In: Proceedings of EMNLP 2016. Austin (TX): ACL; 2016. p. 2383-92. doi:10.18653/v1/D16-12
[29] Tam TYC, Sivarajkumar S, Kapoor S, Stolyar AV, Polanska K, McCarthy KR, et al. A framework for human evaluation of large language models in healthcare derived from literature review. NPJ Digit Med. 2024;7(1):258. doi:10.1038/s41746-024-0125
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[31] Es S, James J, Espinosa Anke L, Schockaert S. RAGAs: automated evaluation of retrieval augmented generation. In: Proceedings of EACL 2024: System Demonstrations. St. Julians: ACL; 2024. p. 150-8. doi:10.18653/v1/2024.eacl-demo.

Method

Agent System Configuration

Input: patient clinical information

AI agent

1 Guideline adherence report

INPUT
Current prescription + clinical information

OUTPUT
How closely the prescription follows each guideline

K-MIMIC Korean (K02, K04) + WHO

MIMIC-IV US (M03, M04, M07) + WHO

2 Guideline-based prescription recommendation

INPUT
Clinical information only

OUTPUT
Recommended empiric regimen with citations

K-MIMIC Korean | WHO

MIMIC-IV US | WHO

3 Similar-case prescription

INPUT
Clinical information only

OUTPUT
Prescriptions given to the most similar past patients

Routing by race / ethnicity → K-MIMIC or MIMIC-IV cases

Data-driven reference, independent of guidelines

Result

MIMIC-IV Antibiotics Prescription Tendency

Pneumonia

4,875
patients

with sepsis
3,666
75.2% · Sepsis-3

Antibiotics Prescription Top 5

vancomycin 50.4 (%)
azithromycin 37.6
ceftriaxone 37.1
cefepime 28.5
levofloxacin 25.5

vancomycin 67.2
cefepime 38.6
azithromycin 30.7
ceftriaxone 29.2
piperacillin/tazobactam 22.2

Urinary Track Infection

4,523
patients

with sepsis
3,180
70.3% · Sepsis-3

Antibiotics Prescription Top 5

ceftriaxone 38.6
ciprofloxacin 24.0
vancomycin 23.9
cefepime 10.3
cefazolin 10.2

vancomycin 47.1
ceftriaxone 35.8
cefepime 21.7
piperacillin/tazobactam 19.0
ciprofloxacin 17.1

Result

MIMIC-IV Antibiotics USA Guidelines Adherence Rate

Pneumonia

4,875
patients

USA guidelines (M03, M04, M07)
Adherence rate

12.5%



adherent 12.5% non-adherent 87.5%

Guideline not complying prescription Top 1 and 2

1. **cefepime + vancomycin**
2. **pip/tazo + vancomycin**

Urinary Track Infection

4,523
patients

USA guidelines (M03, M04, M07)
Adherence rate

39.4%



adherent 39.4% non-adherent 60.6%

Guideline not complying prescription Top 1 and 2

1. **pip/tazo + vancomycin**
2. **cefepime + vancomycin**

Result

K-MIMIC Antibiotics Prescription Tendency

Pneumonia

12,062

patients

Urinary Track Infection

2,895

patients

Antibiotics Prescription Top 10

piperacillin and beta-lactamase inhibitor 19.96 (%)
meropenem 12.81
levofloxacin 9.39
sulfamethoxazole and trimethoprim 6.74
ceftriaxone 4.85
vancomycin 4.52
teicoplanin 4.21
ciprofloxacin 3.61
colistin 2.78
amikacin 2.37

Result

RAG Performance

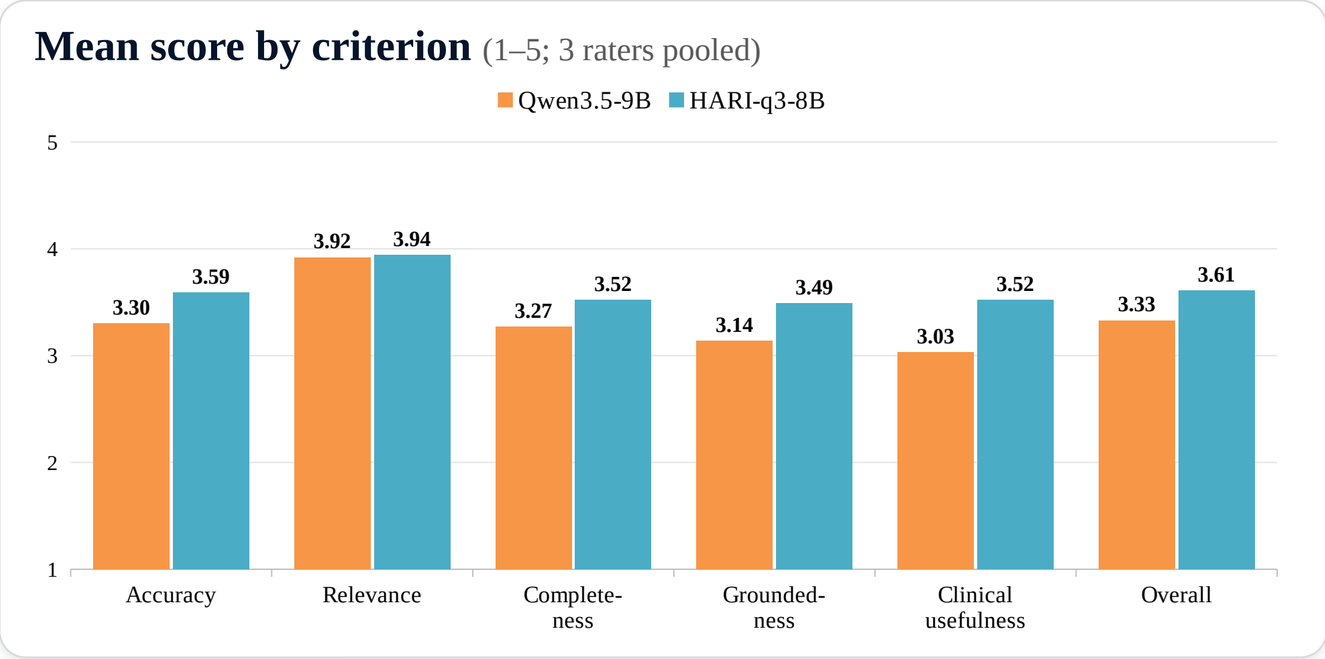
Metric (mean of 21 questions)	Qwen3.5-9B	HARI-q3-8B	HARI better questions
Embedding cosine similarity	0.711	**0.747**	16 / 21
BERTScore F1	0.879	**0.884**	14 / 21
ROUGE-L F1	0.230	**0.253**	12 / 21
Token F1	0.332	**0.351**	12 / 21
'Not found'-type answers	**6 / 21**	**0 / 21**	—

Same embedding (BGE-M3), same 9 retrieved chunks, same prompt — only the generator LLM differs.

Korean medical LLM HARI-q3-8B > general LLM Qwen3.5-9B on every automatic metric

Result

RAG Performance Evaluation By Clinicians



Rater	Role	Qwen3.5-9B	HARI-q3-8B
Seungmin Lee	Neurologist	3.55	**4.03**
Jisu Ha	ICU nurse	3.42	**3.40**
Woojin Kim	ICU nurse, junior	3.03	**3.41**

Question group	K-MIMIC (Korean guidelines)						MIMIC (US guidelines)									WHO (AWaRe)					
	Q01	Q02	Q03	Q04	Q05	Q06	Q07	Q08	Q09	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20	Q21
Qwen3.5-9B	4.7	**1.1**	4.4	4.4	4.5	4.5	4.9	4.2	2.5	2.9	2.9	4.2	2.9	4.8	**1.1**	4.4	2.8	**1.1**	3.0	3.5	**1.1**
HARI-q3-8B	4.3	2.7	4.1	3.7	4.7	4.2	4.9	3.9	3.6	2.8	2.7	3.5	3.4	4.1	3.8	4.6	3.2	3.4	2.4	2.7	3.2

Per-question mean of 5 criteria × 3 raters · colour: **< 2** **2–3** **3–4** **≥ 4**

Agent System Scheme

Limitation

1 Real-time agent deployment was not feasible

- The agent was evaluated on **retrospective case cards**; integrating it with live ICU data streams in real time was difficult to implement

2 Few target diseases, few guidelines

- Only **pneumonia and UTI (± sepsis)** were targeted
- → the number of applicable guidelines was also small (7)

3 Temporal changes in diagnostic criteria

- Changes in diagnostic criteria during the study period could not be fully accounted for, potentially affecting the consistency of patient classification
- → **past data were judged against recent guidelines**

Conclusion

Key results

1 Guideline RAG in two LLM versions

Qwen3.5-9B vs HARI-q3-8B, assessed with automatic metrics and by 3 clinicians

2 AI agent built around clinicians' needs

adherence report · guideline-based recommendation · similar-case prescription

3 Antibiotic tendency in MIMIC-IV & K-MIMIC

MIMIC-IV: top-5 by disease & sepsis · K-MIMIC: top-10 (pneumonia + UTI)

Further research

- ▸ **External & prospective validation**
  other hospitals, real-time ICU workflow
- ▸ **Verification agent**
  check every claim against cited text to curb hallucination
- ▸ **Broader populations**
  HAP/VAP, intra-abdominal, skin & soft tissue; ED and wards
- ▸ **Outcome linkage**
  mortality, resistance, C. difficile; local antibiogram
- ▸ **Better retrieval**
  source routing, reranker, table-aware chunking
- ▸ **Robust expert evaluation**
  more blinded raters, κ / ICC, agent vs clinicians

Thank You